Exam Date & Time: 03-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

V SEMESTER B.TECH END SEMESTER MAKEUP EXAMINATION, DEC 2024 - JAN 2025

ARTIFICIAL NEURAL NETWORKS [BME 3124]

Marks: 50 **Duration: 180 mins.**

Descriptive

Answer all the questions.

Instructions to Candidates:
[i] Missing data may be suitably assumed
[ii] No formula book will be provided

1A) Using the gradient descent algorithm, design a binary classifier for a dataset with two features x_1 and x_2, where the initial weights and bias are set to small random values, and the learning rate is 0.05. Describe the steps required to train this classifier, including how the weights and bias are updated through gradient descent. (5)

1B) For a multi-class classification problem, the predicted probabilities for a sample are [0.7, 0.2, 0.1] with the true label being the second class. Calculate the categorical cross-entropy loss for this prediction. Explain what the loss value indicates about the accuracy of the model's prediction and how this metric helps in optimizing network weights. (3)

1C) A neural network applies dropout at a rate of 0.5 and achieves a test accuracy of 93%. If the dropout rate is reduced to 0.2, the test accuracy drops to 89%. Calculate the difference in test accuracy and interpret how dropout rate quantitatively influences model generalization. (2)

2A) In the backpropagation process, consider a three-layer network with loss calculated at the output layer. (5)

i. Describe how error propagation occurs back through the hidden layers

ii. Explain the importance of adjusting weights and biases based on this propagated error and how this process optimizes the network's performance over multiple iterations.

2B) A model is trained with batch sizes of 16, 32, and 64, achieving validation accuracies of 85%, 88%, and 92%, respectively. Explain how increasing batch size can impact model stability and convergence speed, both quantitatively and qualitatively, hypothesize what batch size might yield the optimal result without causing memory constraints. (3)

2C) Describe the significance of avoiding self-connections in a Hopfield network. Explain (2)

how setting $W_{ii} = 0$ for each neuron impacts the network's retrieval ability.

3A) Explain how a neuron's state x_i is updated in a Hopfield network. Suppose the weighted sum of inputs for a neuron is 3. What will the updated state of this neuron be? (3)

3B) Compare the effects of using RNN and LSTM models for predicting future values in a sine wave. Analyze how the internal structure of LSTMs (such as the cell state and forget gate) helps them retain information over longer sequences compared to simple RNNs. (3)

3C) Analyze the role of the update and reset gates in a GRU for time-series prediction. How does this simplified structure compare to LSTMs in terms of computational efficiency and handling dependencies? (4)

4A) Consider a CNN model with multiple convolutional layers that result in 64 feature maps of size 4x4, which are then flattened before being passed into a fully connected layer with 128 neurons.

i.Determine the size of the input vector for the fully connected layer. (4)

ii. Calculate the number of weights in the fully connected layer.

4B) When building a CNN for object detection, two activation functions are considered for the convolutional layers: ReLU and sigmoid.

i. Choose the more appropriate activation function for a CNN's convolutional layers and explain why. (3)

ii. Analyze the potential drawbacks of using the alternative function in this context.

4C) Explain how max pooling with a 2x2 window and stride of 2 reduces overfitting in CNNs. Discuss its impact on the spatial dimensions of feature maps. (3)

5A) For a digital pathology task, design a data augmentation strategy to improve a CNN model's robustness in detecting cancerous cells. In your answer, include the following:

i. List the specific augmentation techniques you would apply to histopathology slides and justify each. (4)

ii. Explain how the proposed strategy reduces model sensitivity to variations in cell morphology and staining.

5B) When preparing EEG data for training, two preprocessing options are available: (3)

Option A: Applying bandpass filtering and segmentation into 5-second windows.

Option B: Using only data segmentation without additional filtering.

i. Analyse and select the option that will likely yield better results for seizure prediction and justify your choice.

ii. Explain how the preprocessing technique chosen affects the neural network's ability to detect relevant features in EEG signals.

5C) You are tasked with training a binary classifier to differentiate normal and abnormal heart rhythms in ECG data. Given the following configurations: (3)

Option A: Small feature vector (10 features) with one hidden layer in an FNN.

Option B: Large feature vector (50 features) with two hidden layers in an FNN.

i. Choose the configuration that balances model complexity and performance for this binary classification task.

ii. Justify your choice and explain the potential impact on overfitting and computational load.

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